Ardalan Siavashpour

ardalan.siavashpour@gmail.com | github.com/Ardalan-Sia

EDUCATION

Sharif University of Technology, Tehran, Iran *Nov. 2020 – Jul. 2026*
Bachelor of Science in Computer Science; graduated July 2026

RESEARCH INTERESTS

Machine learning for speech and multimodal information retrieval; reinforcement learning and sequential decision-making; intelligent networked systems.

RESEARCH EXPERIENCE

Undergraduate Researcher – Speaker Recognition *2025 – Present*
First author · Supervisor: Dr. Mohammad Javadian

- Implemented block-based audio preprocessing and MFCC feature extraction with first- and second-order derivatives, spectral centroid, spectral roll-off, and zero-crossing rate.
- Applied ANOVA F-test feature selection and compared multilayer perceptron and linear SVM classifiers using a shared feature pipeline and cross-validation for feature-count and SVM hyperparameter selection.
- Led data preparation, model development, evaluation, and manuscript writing; currently revising the manuscript for a new submission.

Undergraduate Research Assistant – Self-Organizing Networks *Summer 2024 – Present*
Sharif University of Technology · First author
Supervisors: Dr. Hossein Peyvandi and Dr. Masoumeh Azadegan

- Co-authored a survey synthesizing 114 references on self-configuration, self-optimization, self-healing, self-learning, and self-protection in 4G/5G and emerging 6G cellular systems.
- Analyzed SON architectures, closed-loop control, performance objectives, and control parameters; compared optimization-based, heuristic, and learning-driven methods in terms of scalability, adaptability, and robustness.
- Co-wrote major manuscript sections and developed comparative tables and technical figures in LaTeX and TikZ; examined open challenges in function coordination, explainable and federated learning, and non-terrestrial network integration.

RESEARCH MANUSCRIPTS

- **A. Siavashpour**, S. Rajabi, and M. Javadian, "Speaker recognition based on Mel-frequency cepstral coefficients using machine learning methods." *Manuscript being revised for a new submission.*
- **A. Siavashpour**, K. Siavashpour, M. Azadegan, B. Safaei, H. Peyvandi, and M. Imran, "A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems." *Manuscript in preparation.*

TEACHING EXPERIENCE

Teaching Assistant, Computer Networks *Spring 2026; Fall 2026 – Present*
Sharif University of Technology · Instructor: Dr. Mahdi Dolati

- Responsible for selected theoretical exercises and the course project in Fall 2026.
- Designed the final course project and its evaluation criteria, and evaluated student submissions in Spring 2026.

Teaching Assistant, Software Engineering *Fall 2025*
Sharif University of Technology · Instructor: Dr. Mehran Rivadeh

- Led weekly recitations on design patterns, development methodologies, and software architecture; evaluated team projects and guided testing and version-control workflows.

Teaching Assistant, Computer Networks *Spring 2024*
Sharif University of Technology · Instructor: Dr. Hossein Peyvandi

- Conducted recitations and labs; developed problem sets, exams, and grading rubrics; maintained automated grading and prepared materials on TCP congestion control, IP routing, and transport-layer reliability.

Teaching Assistant, Computer Networks *Fall 2023*
Sharif University of Technology · Instructor: Dr. Kambiz Mizanian

- Developed networking problem sets and programming exercises and graded examinations and assignments.

SELECTED PROJECTS

Torob Agentic RAG *2026*
[GitHub]

- Built a Persian e-commerce retrieval pipeline indexing 200,000 products with multilingual embeddings, FAISS search, and lexical reranking; integrated CLIP image retrieval and text–image fusion.
- Developed a conversational shopping-agent HTTP API with conversation history, preference memory, runtime metrics, and automated smoke tests.
- Evaluated text retrieval on 5,000 search-log queries using Recall@k, MRR, and NDCG; analyzed retrieval errors and limitations of product-subset coverage.

Job-Shop Scheduling Algorithms *Spring 2024*
[GitHub]

- Co-implemented classical and learning-based scheduling approaches, including a Johnson-inspired heuristic, genetic algorithms, and Deep Q-Learning.
- Compared makespan, waiting time, and response time across scheduling approaches in a simulation framework.

SliceDB – Distributed Key-Value Database *2026*
[GitHub]

- Co-developed a distributed in-memory key-value store in Go with hash-based partitioning, asynchronous leader/follower replication, automatic failover, and online resharding.
- Contributed to etcd-based service discovery, lease-based failure detection, controller leader election, and fenced routing-table updates.
- Worked on write-ahead log (WAL) replication, snapshot-based replica synchronization, replica balancing, and dynamic cluster reconfiguration.

Highly Available HDFS and Spark Analytics *2025*
[GitHub] · Course project

- Co-built a Dockerized HDFS deployment with two NameNodes in an active/standby configuration, three DataNodes, three JournalNodes, ZooKeeper, and automatic NameNode failover.
- Configured a logical HDFS nameservice for failover-aware clients and developed Java Spark Dataset jobs analyzing NYC taxi records and writing four query outputs as Parquet.

Online Judge Platform *Spring 2025*
[GitHub]

- Co-developed a Go programming-judge platform with role-based access, problem management, submission tracking, and asynchronous verdict generation using Gin and PostgreSQL.
- Built a separate code-runner service that executes Go submissions in Docker containers with CPU, memory, process-count, and network restrictions; compares output against the problem's sample test.

TECHNICAL SKILLS

Programming: Python, Go, C++, Java, R
Machine Learning: TensorFlow/Keras, scikit-learn, neural networks, SVM, reinforcement learning, speech features, text and image retrieval
Tools & Systems: FAISS, CLIP, Docker, Git, Linux, PostgreSQL, Apache Spark, Hadoop HDFS, LaTeX, TikZ

LANGUAGES

Persian: Native
English: IELTS Academic 7.0 overall (Listening 8.0, Reading 7.0, Writing 6.5, Speaking 6.5; December 2025)

HONORS & AWARDS

Top 0.5% nationwide, Iranian University Entrance Examination (Konkoor) *2020*

First Place, Allameh Tabatabaei Schools Nanotechnology Competition *2018*